

Smart Water Level Monitoring System

Project Report | Harsh Kumar | B.Tech CSE (AI & ML), Lovely Professional University | 2025

1. Abstract

The Smart Water Level Monitoring System is an Arduino-based embedded project designed to monitor the water level inside a tank or container in real time. A water level sensor detects the current water level and sends analog readings to the Arduino for processing. The system provides both visual (LED) and audio (buzzer) indications as the water level changes, and displays live status on a 16x2 LCD screen. The primary goal of the project is to prevent tank overflow and reduce unnecessary water wastage through automated, low-cost monitoring.

2. Components Used

- Arduino Uno (microcontroller board)
- Water Level Sensor Module
- 16x2 LCD Display with I2C interface module
- 3x LEDs (Red, Yellow, Green) for Low / Medium / Full indication
- Buzzer for full-tank alert
- Resistors, breadboard, and jumper wires

3. Working Principle

The water level sensor is placed inside the tank and continuously outputs an analog voltage proportional to how much of the sensor is submerged in water. This value is read by the Arduino's analog input pin and converted into a 0-100% water level reading. Based on this percentage, the system classifies the tank status as Low, Normal, or Alert, lighting up a corresponding LED. The live percentage and status are shown in real time on the LCD display (e.g. "Water Level: 67% - Status: Normal"). When the level crosses the Alert threshold, the buzzer is triggered to warn the user, helping avoid overflow.

4. Circuit Overview

The water level sensor's analog output is connected to pin A0 of the Arduino. The Red, Yellow, and Green LEDs are connected (through current-limiting resistors) to digital pins 8, 9, and 10 respectively. The buzzer is connected to digital pin 7. The LCD communicates with the Arduino over I2C (SDA/SCL pins), reducing the number of wires needed compared to a standard parallel-interface LCD.

5. Arduino Code

Full source code is included in the project repository (smart_water_level_monitor.ino). Key logic snippet:

```
int rawValue = analogRead(sensorPin);
```

```
int percent = map(rawValue, emptyReading, fullReading, 0, 100);
percent = constrain(percent, 0, 100);

if (percent < 30) {
    digitalWrite(redLED, HIGH);
    status = "Low";
}
else if (percent < 90) {
    digitalWrite(yellowLED, HIGH);
    status = "Normal";
}
else {
    digitalWrite(greenLED, HIGH);
    digitalWrite(buzzerPin, HIGH);
    status = "Alert!";
}

lcd.print("Water Level: " + String(percent) + "%");
lcd.print("Status: " + status);
```

6. Results

The system successfully detected and displayed four distinct water level states in real time, with the LED indicators and buzzer responding correctly at each threshold. The LCD provided a clear, constantly updated readout of tank status, confirming the system's reliability for practical use in homes, overhead tanks, and small-scale irrigation setups.

7. Conclusion

This project demonstrates a practical application of embedded systems and sensor interfacing using Arduino. It highlights core skills in circuit design, analog signal processing, and real-time hardware-software integration — offering a low-cost, scalable solution to a common real-world problem of water overflow and wastage.